

ISO-1 weak isogeny classes: trace census and invariant audit

Research evidence report

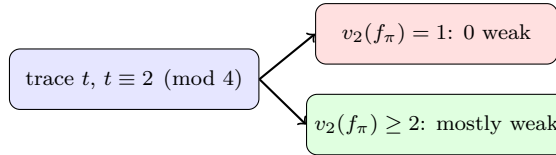
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Requested scope. Base primes $p = 37$ through about 200, $q = p^2$, and curves over $\mathbb{F}_{q^3} = \mathbb{F}_{p^6}$: label every full-2-torsion isogeny class and find a criterion for weak-class reach. This round completed the every-trace census at $p = 37$ and diagnostic censuses at $p = 11, 13, 17$. The other requested primes and an exact necessary-and-sufficient criterion remain open.

Method. A weak representative has $y^2 = x(x - \alpha)(x - \sigma\alpha)$, with σ the p^2 -Frobenius. After translation and $\mathbb{F}_{p^2}$ square scaling there are $2q^2 + 2q$ normalized representatives. The $p = 37$ run evaluated one representative per quadratic-twist pair and derived the opposite trace. Trace assignment uses a randomized two-point baby-step/giant-step counter. Its labels are computational evidence, not independent certificates of every cardinality. Full and twist-derived CSVs agreed exactly at $p = 13$; two false individual assignments at $p = 7$ were found by exact square-table counting.

p	q	weak reps	ordinary t	weak t	depth-1 weak	depth- ≥ 2 zero
11	121	29,524	1,210	542	0/606	62/604
13	169	57,460	2,028	928	0/1,014	86/1,014
17	289	167,620	4,624	2,198	0/2,312	114/2,312
37	1,369	3,751,060	49,284	24,352	0/24,642	290/24,642

At $p = 37$ the run took 1,106.807 seconds with four Rayon threads and counted 563,856,133,786 $\mathbb{F}_p$ multiplications. The 50,654 CSV rows cover every candidate $t \equiv 2 \pmod{4}$ in the Hasse interval; 49,284 are ordinary. Of 4,000 sampled random full-2-torsion curves, 2,433 were in an observed weak class: 0.6083, Wilson 95% interval [0.5930, 0.6233]. The zero-label share is 50.59% of ordinary trace rows but 39.18% of sampled curves; these use different denominators.



Candidate invariant. Let $t^2 - 4p^6 = f_\pi^2 D_K$, where D_K is the fundamental CM discriminant. The measured $v_2(f_\pi)$ is a class-wide bound, not a certified endomorphism-ring volcano level. For ordinary $t \equiv 2 \pmod{4}$, the arithmetic tests

$$v_2(f_\pi) \geq 2 \iff (t/2)^2 \equiv p^6 \pmod{16} \iff t/2 \equiv \pm p^3 \pmod{8}$$

are equivalent. All observed weak classes at $p = 11, 13, 17, 37$ pass. This is an *empirical necessary condition*; universal necessity is unproved here. It is not sufficient. On held-out $p = 37$ rows it has 24,352 true positives, 290 false positives, no false negatives, and 24,642 true negatives (99.41% correct).

Counterexample to the proposed small invariants. At $p = 37$, $t = -92,218$ has zero observed weak representatives and $t = 38,854$ has 24. Both have Frobenius conductor depth 3, ramified 2, even maximal-order class number, and the same trace modulo 2^{17} . Their PARI/GP class numbers are 912 and 3,168. Thus 2-splitting, conductor depth, small 2-power trace residues, and class-number parity do not classify all rows together.

Where the residual zeros occur. At $p = 37$, 234 of 290 high-depth zero rows lie in the outer fifth of $|t|/(2p^3)$ (234/4,930 rows there, versus 6/4,926 in the inner fifth). This is an association, not a causal proof or an exact rule.

Correction to prior evidence. The earlier census used an $\mathbb{F}_p$ nonsquare for the second $\mathbb{F}_{p^2}$ square class. Every nonzero element of $\mathbb{F}_p$ is a square in $\mathbb{F}_{p^2}$, so it duplicated one branch and omitted the other. The corrected code selects a genuine $\mathbb{F}_{p^2}$ nonsquare. Old $p \leq 31$ values remain historical; $p = 23, 31$ await corrected reruns.

Limit. At $p = 199$ the same algorithm would need 1,568,278,802 direct point counts even after twist symmetry. Scaling the measured $p = 37$ run by work count and the approximate $p^{3/2}$ cost per point count gives roughly 130 days on comparable resources for that prime alone. This is an extrapolation, not a run. The requested $p = 41$ –200 census and a replacement formula have not been established.

Evidence. The accompanying Markdown report links the code, complete CSVs, fit output, samples, and SVG. The $p = 37$ command used `RAYON_NUM_THREADS=4` and `--derive-twists`. The weak-model and reach assumptions are in [Joux–Vitse, §4.1](#); [Auer–Top](#) proves a broader Legendre-family statement.